

Introduction to Java Web Application Development (PRJ301)

Kiều Trọng Khánh

<https://lms-hcm.fpt.edu.vn/>
KhanhKT

Prerequisites

- **Completed:**
 - DBI202 (Database Systems)
 - PRO192 (Object Oriented Programming)

Course Objectives

TEACHING METHODOLOGY

Project Based Learning (PBL)

Học qua dự án thực tế



COURSE OBJECTIVES

Explore JavaEE/J2EE features

Develop Java Web App (BackEnd)

Apply MVC Design Pattern

Use Servlet/Filter as Controller



CORE TECHNOLOGIES

Servlet JSTL
JDBC EL
JSP JPA
JSTL Filter
(Not Framework Usage)



DEVELOPMENT TOOLS

JDK 8 (Update 172)
Apache Netbeans 13
Apache Tomcat 9.0.113
MS SQL Server 2019



DURATION

Total: **10 weeks**

Implementation: **8 weeks**

Practical Exam & Final Assessment: **2 weeks**



ARCHITECTURE

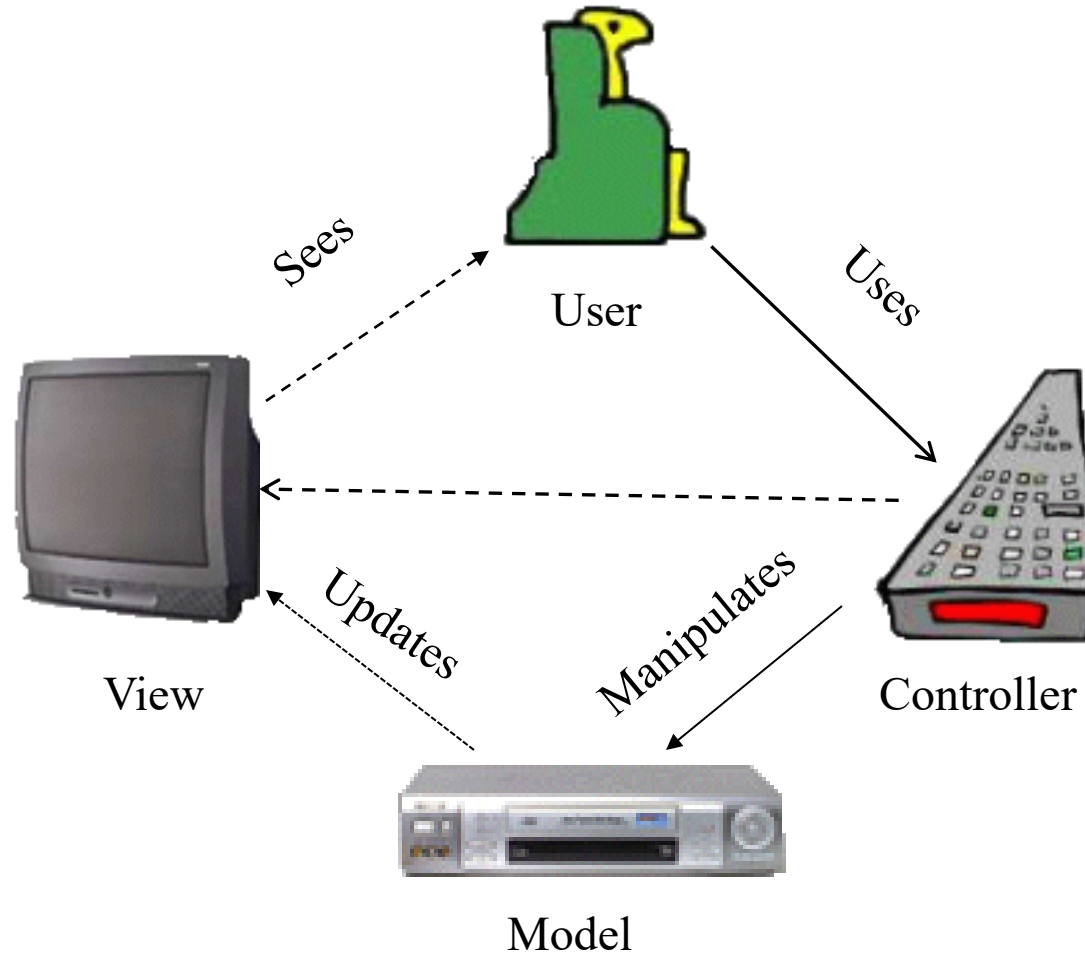
MVC2 Design Pattern

Servlet/Filter as Dispatcher/Main Controller



MVC Design Pattern

Model – View – Controller



This is a MVC Model

Course Objectives

- The MVC architecture is applied in web application



Course Objectives

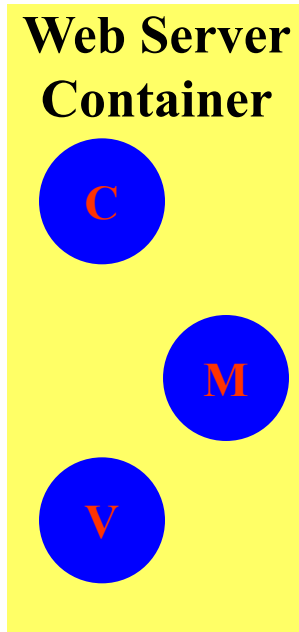
- The MVC architecture is applied in web application

**Web Server
Container**



Course Objectives

- The MVC architecture is applied in web application

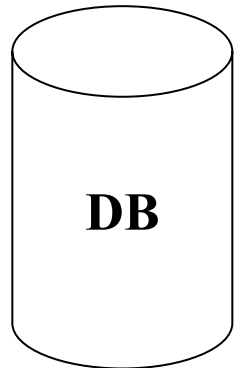
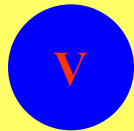
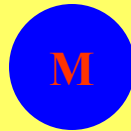
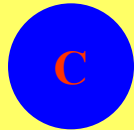


Course Objectives

- The MVC architecture is applied in web application

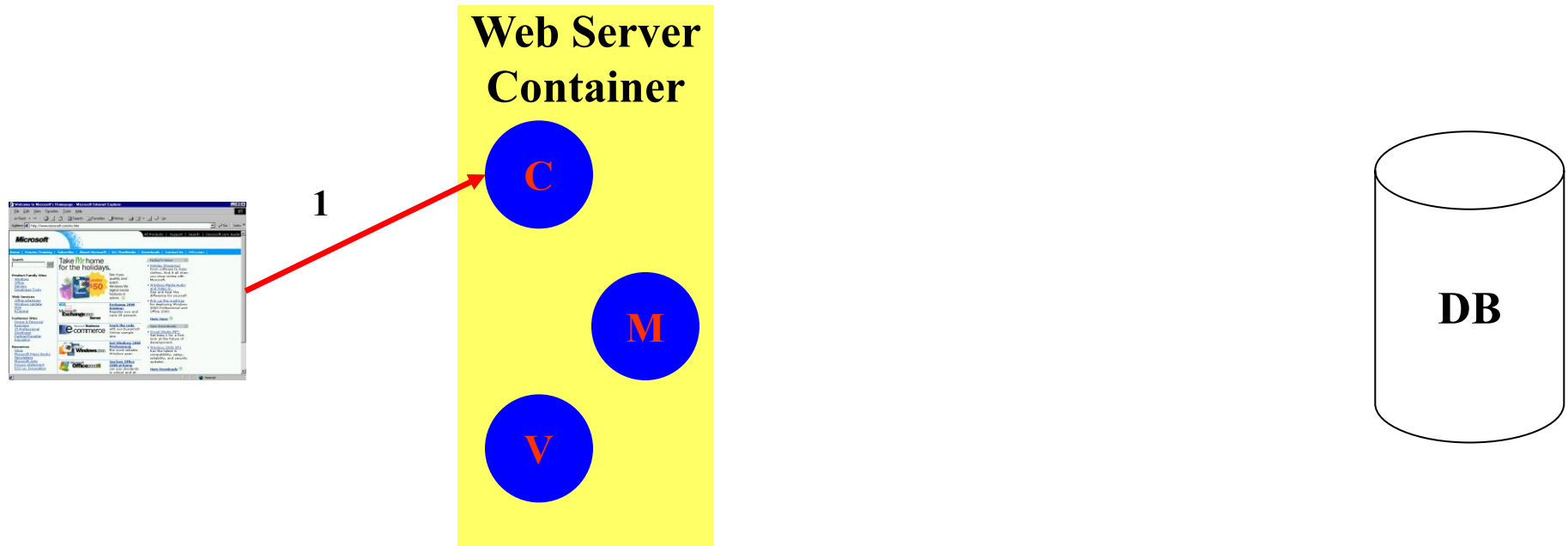


**Web Server
Container**



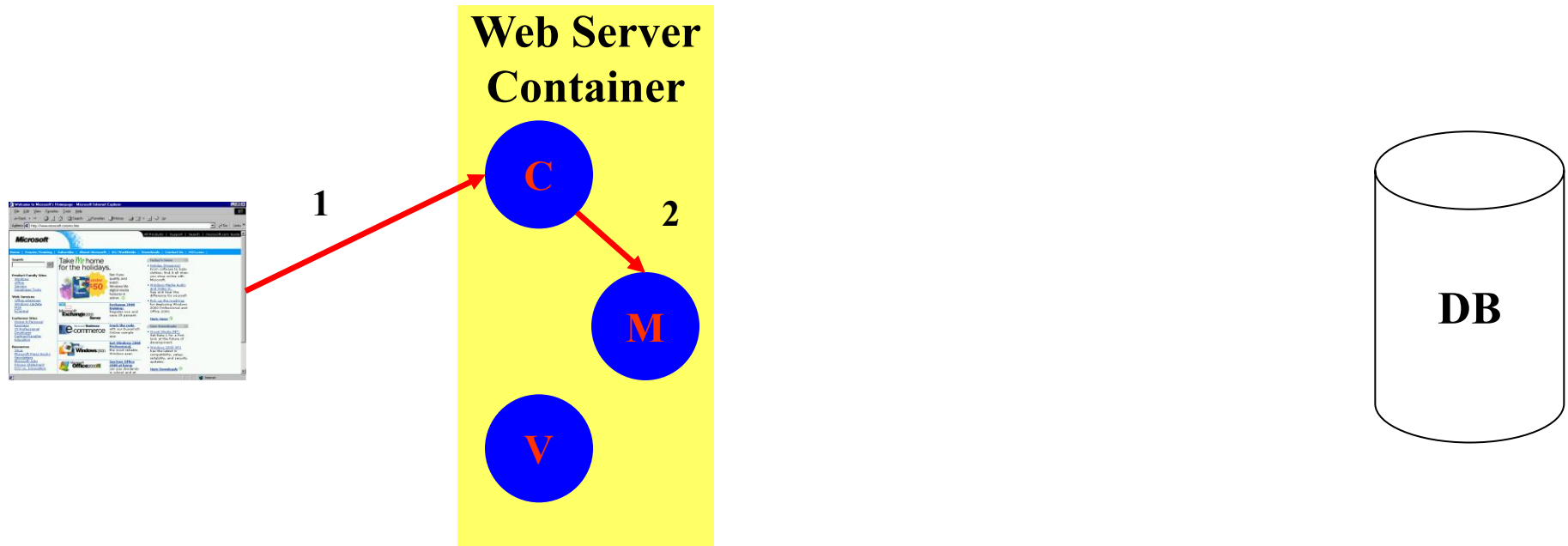
Course Objectives

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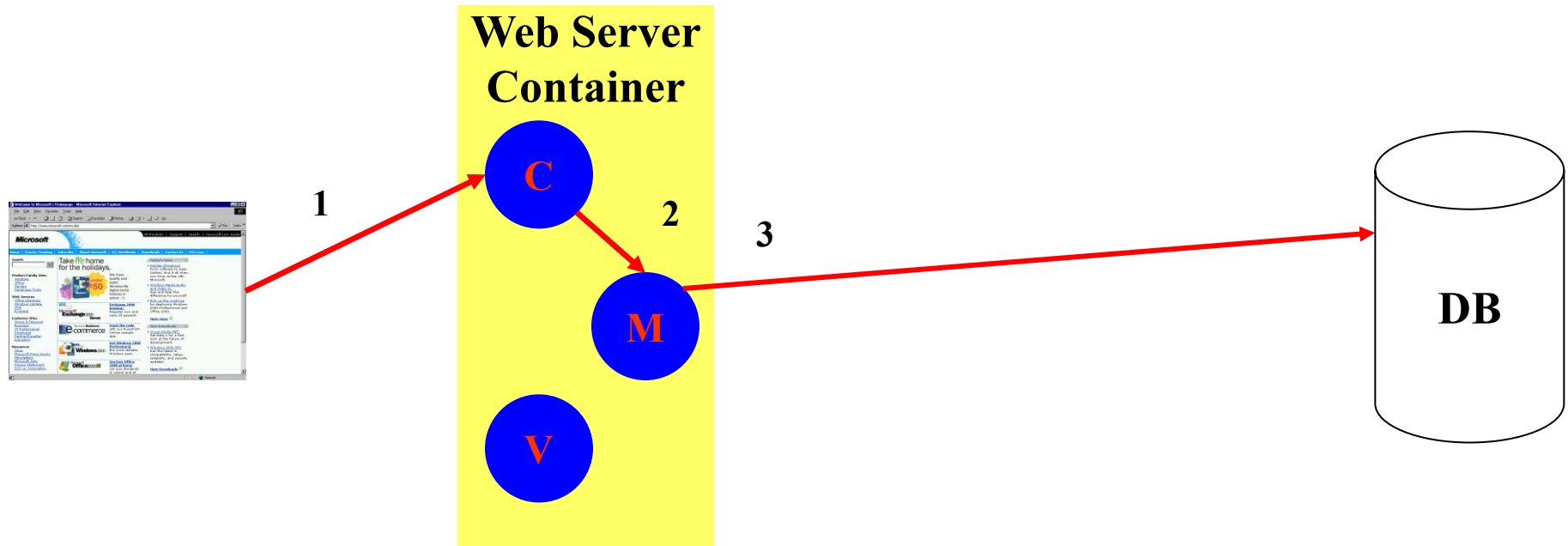
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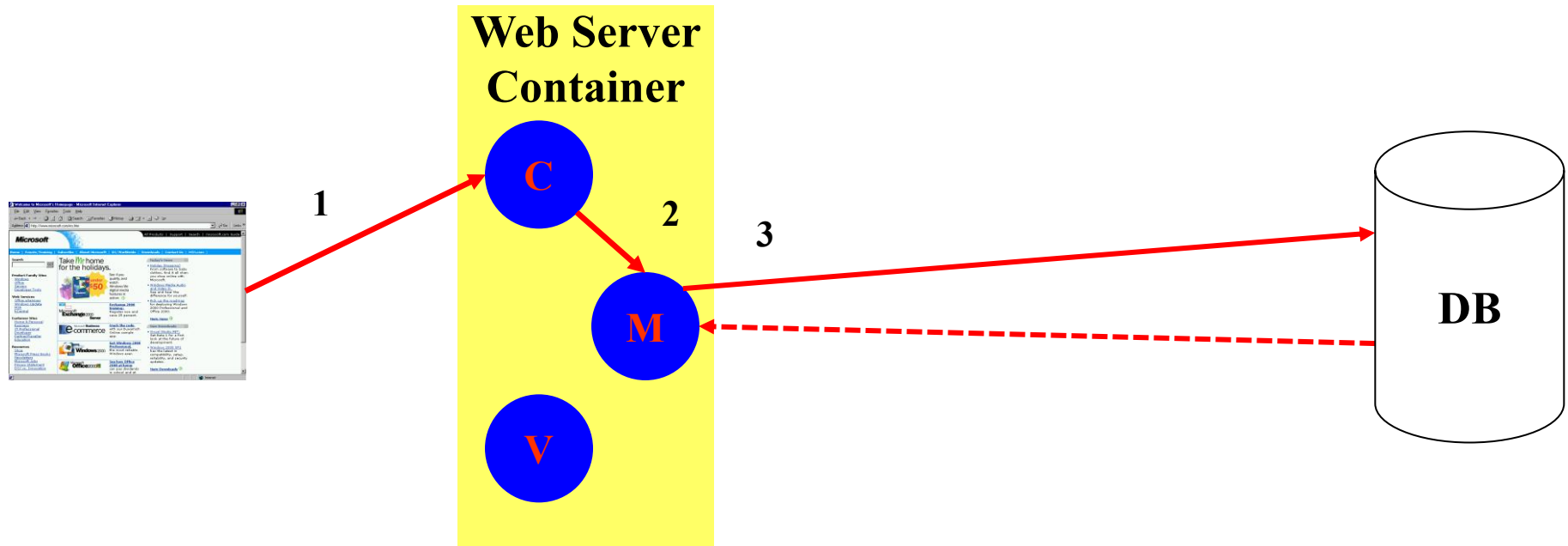
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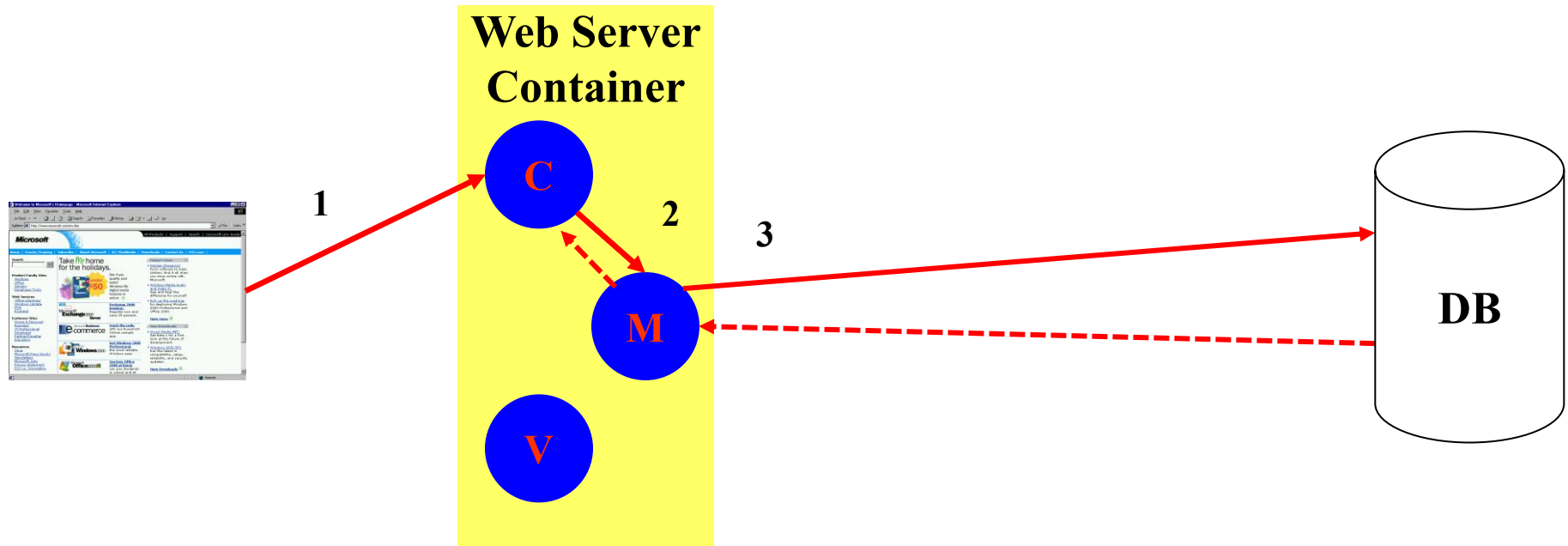
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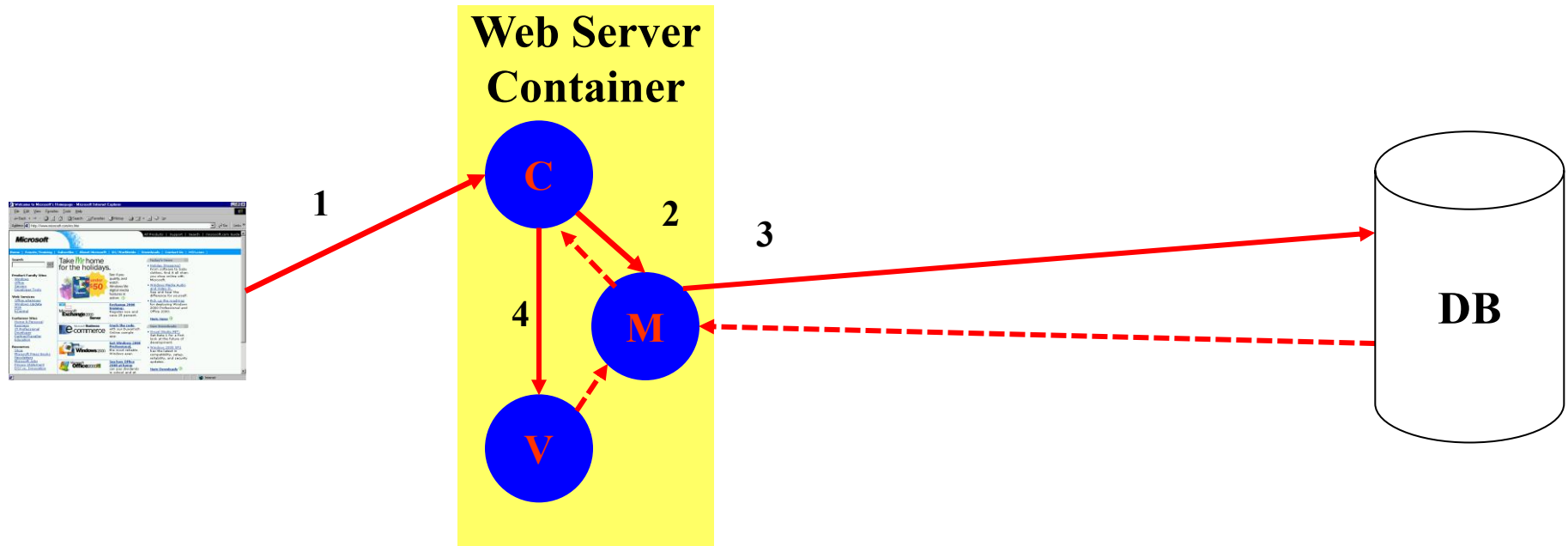
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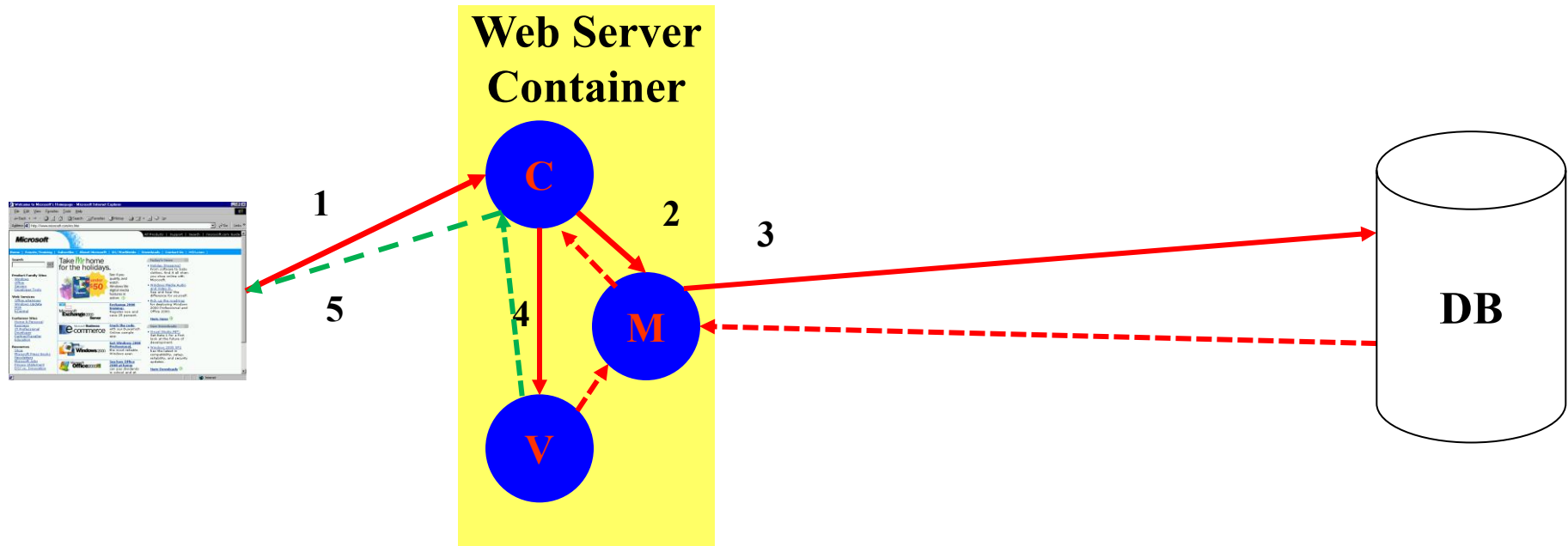
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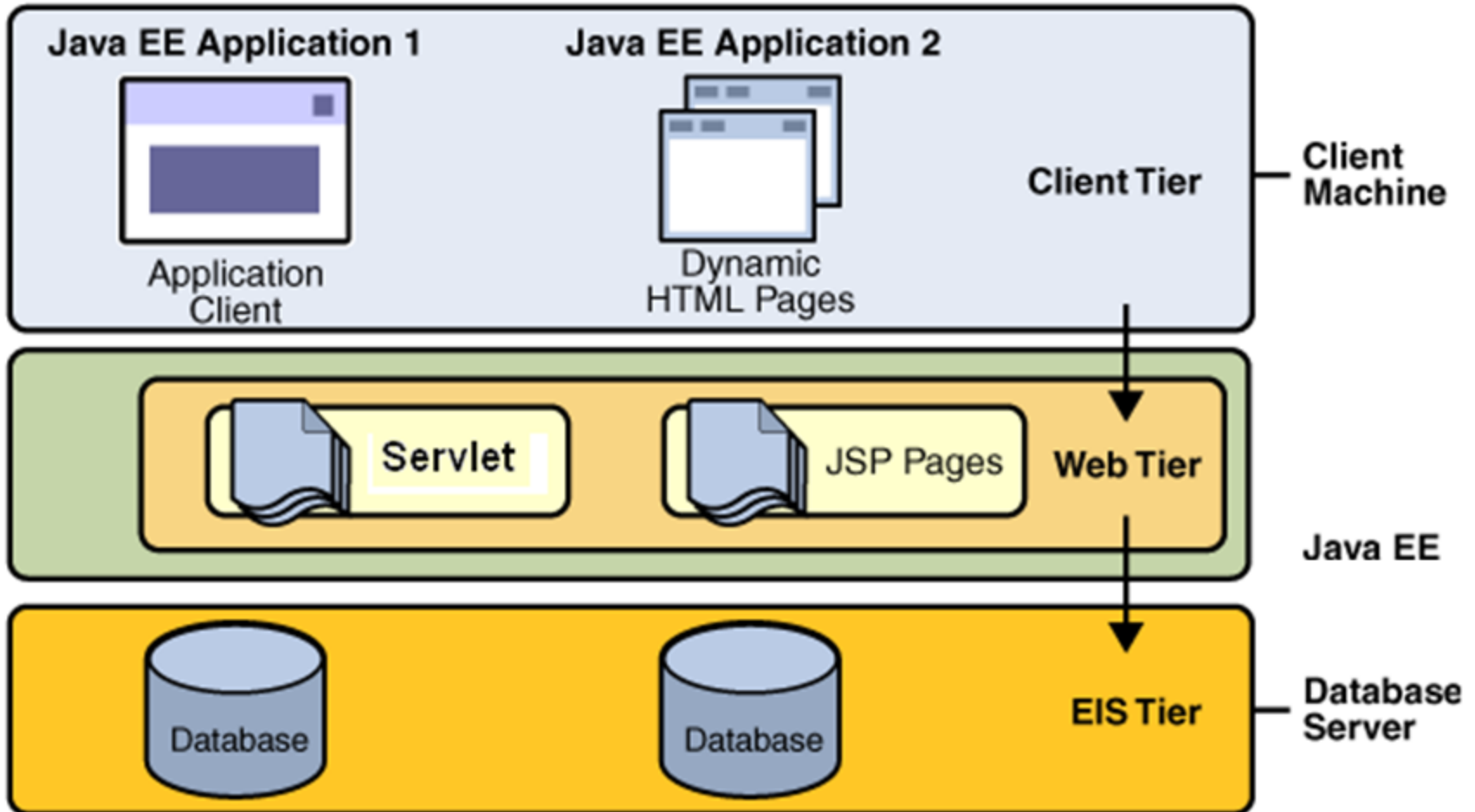


Course Objectives

- The MVC architecture is applied in web application

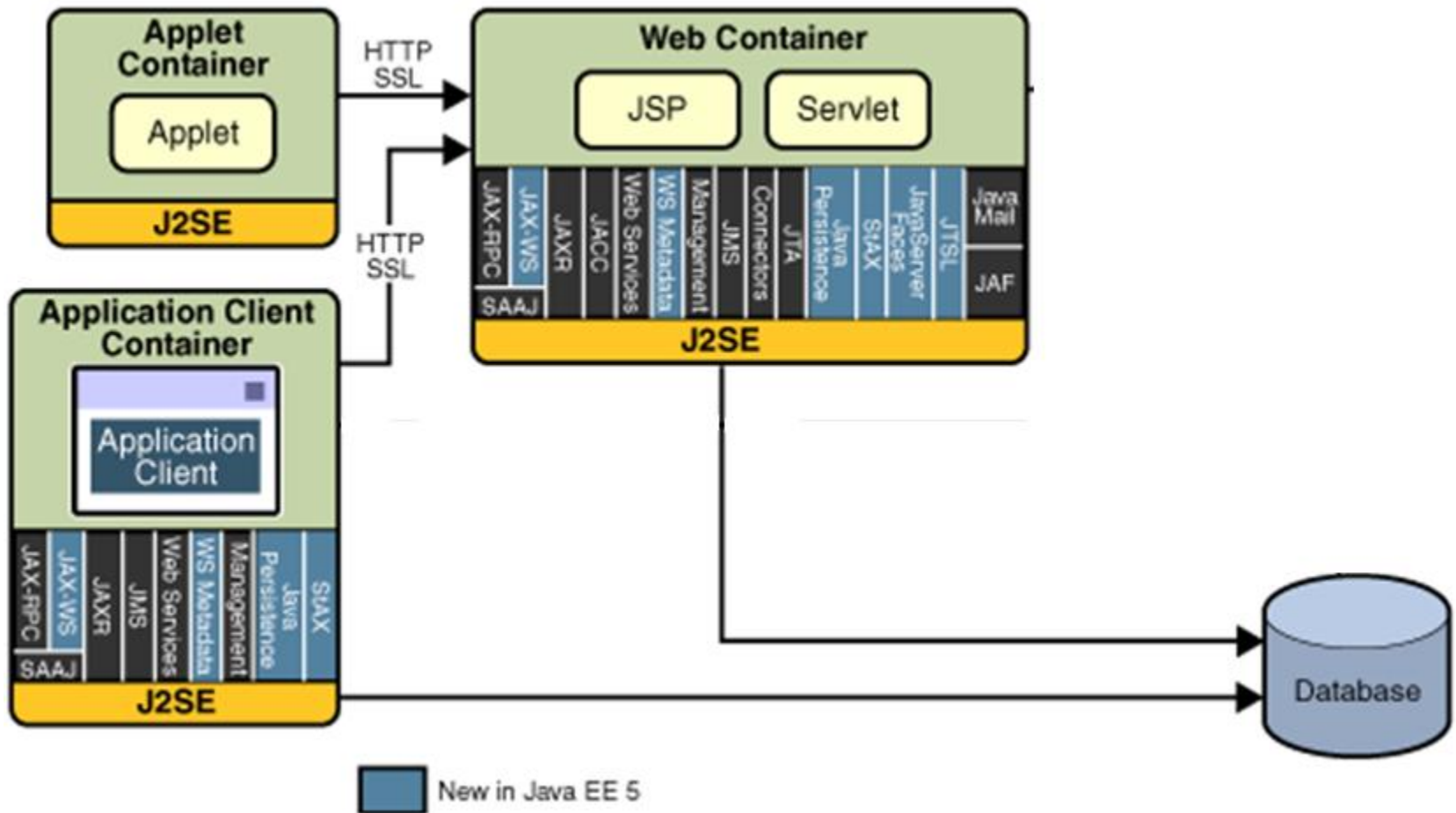


Course Objectives



Course Objectives

- **J2EE 1.4/ JavaEE5/ JavaEE6 Platform API**



Course Plan

- See course plan on LMS/EduNext
- Agenda
 1. The Servlet Model
 2. Web Application Interacting with Database
 3. Web Application & Web Container
 4. Java Server Pages (JSP)
 5. Session Management, Session & Listener (self-study)
 6. JSP Tag Libraries
 7. JPA – Java Persistence API
 8. Filters
 9. JPA
 10. *Practical test*
 11. *Project Presentation*

Course Plan

- See course plan on LMS/EduNext



Software Engineering Web Application Timeline



Week 1

Initialization & Core Technologies

- MVC2 & DB Setup
- Login UI Creation
- Deployed on Tomcat.



Week 2

Servlet & HTTP Request

- GET/POST Handling.
- Login Handling.
- Login Servlet Success.



Weeks 3–4

JDBC, DAO & Business Logic

- JDBC Connection
- Controller Setup.
- DB & Console Results.



Week 5

JSTL & Rendering

- JSP & JSTL Use
- Data Grid Display.
- MVC2 Separation.



Weeks 6–7

Session & CRUD

- Create, Update, Delete.
- Session Management.

Week 8

JPA & Management

- JPA Migration.
- User & Product.

Week 9

Filter & Listener

- Filter Dispatcher.
- Resource Mapping.

Week 10

Finalization & Demo

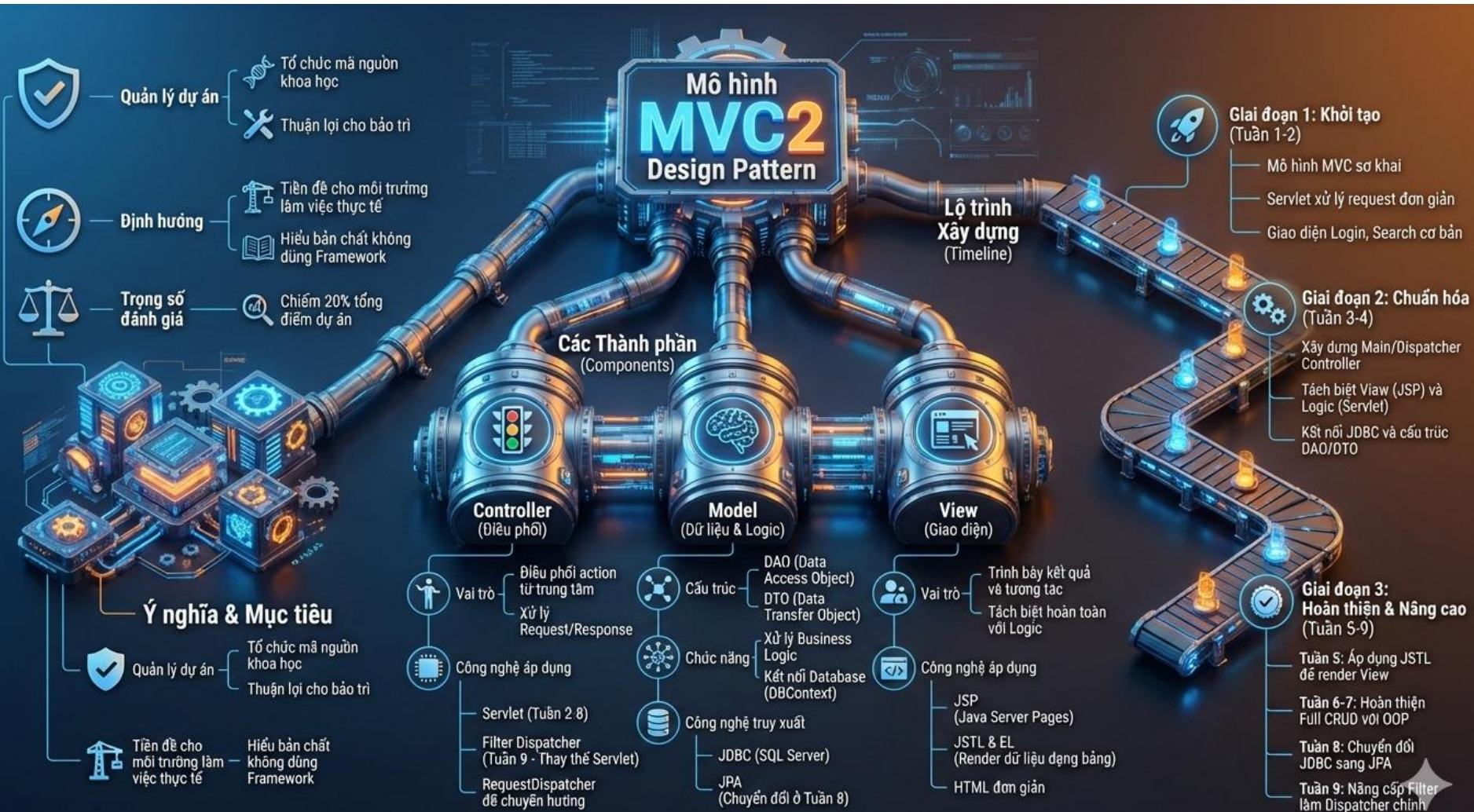
- Practical Exam.
- System Demo.

Course Plan

- See course plan on **LMS/EduNext**

Week	Main Topic	Technical Objectives & Tasks	Expected Deliverables
Week 1	Initialization & Core Technologies	- Understand MVC2 model and Web Application architecture. - Create SQL database and initialize project skeleton. - Implement Login UI (interface only).	- Project successfully deployed on Tomcat. - HTML pages: Login, Search, Invalid.
Week 2	Servlet & HTTP Request/Response	- Handle GET/POST methods and URL mapping. - Receive parameters from Login page. - Apply DAO pattern to separate logic.	- Servlet processes successful Login request. - Display received username/password on browser.
Weeks 3–4	JDBC, DAO & Business Logic	- Establish database connection using JDBC. - Construct Main Controller / Dispatcher Controller. - Implement real Login and Search functionality (approximate matching).	- Successful DB connection and data retrieval. - Login and search results displayed in console/log.
Week 5	JSTL & Rendering	- Use JSP as View layer. - Apply JSTL/EL for data presentation.	- Search results displayed as Data Grid (table). - Ensure MVC2 applying correctly components.
Weeks 6–7	Session Tracking & Full CRUD	- Complete database operations: Create, Update, Delete. - Manage user sessions - Apply OOP thoroughly in DAO and DTO.	- Full CRUD functionality: Add, Update, Delete products based on Primary Key.
Week 8	JPA & Product Management	- Migrate data layer from JDBC to JPA. - Manage User and Product entities.	- Source code for User/Product management using JPA.
Week 9	Filter & Listener	- Replace Servlet Dispatcher with Filter-based Dispatcher. - Enable flexible resource mapping.	- Project reconstructed with Filter as Main Controller.
Week 10	Finalization & Demo	- Prepare for Practical Exam. - Finalize all system functionalities.	- Practical exam conducted at the end of Week 9. - Project evaluation and full system demo.

Course Objectives



Materials/ References

- **Required Textbook**

- Online Text book: Nicholas S. Williams, 2014, Professional Java® for Web Applications, Wrox Press
- <http://library.books24x7.com/toc.aspx?bookid=62587>

- **Required References**

- <http://java.sun.com/docs/books/tutorial/jdbc/>
- https://docs.oracle.com/cd/B14099_19/web.1012/b14017/filters.htm

- **References**

- <http://java.sun.com/>
- Reference Page: <https://www.kieutrongkhanh.net/>
- Fan Page: <https://www.facebook.com/TrongKhanh.Kieu/>

Learning Environments

- JDK 8 (<https://www.oracle.com/java/technologies/downloads/>)
 - **Recommend:** JDK 8 Update 172
(<https://www.oracle.com/java/technologies/javase/javase8u211-later-archive-downloads.html>)
- JDK 8 Documentation
- J2EE 1.4/JavaEE7 Core Patterns
- **NetBeans IDE 13 with supporting JavaEE**
(<https://www.apache.org/dyn/closer.lua/netbeans/netbeans-installers/13/Apache-NetBeans-13-bin-windows-x64.exe>)
- **Tomcat 9.0.113** (<https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.113/bin/apache-tomcat-9.0.113-windows-x64.zip>)
- **DBMS: MS. SQL Server 2019** (<http://www.microsoft.com/sqlserver>)
- **Browser: Edge** (should have **Internet Explorer simulator**)
- **Driver Type 4 for MS. SQL Server: sqlserver4.jar**
(on LMS)
- **GG Meet for supporting (efc-padj-aob)**

Course Rules



How to conduct

Prepare contents of the next session/ topic at home

Following lessons in classrooms (take note yourself to understand using notebooks) and review previous session in lecture on class every day (penalty marks on workshops)

At least 02 session for studying social constructivism with deposit for Assignment

Completing chapter assessments in time and Quizzes (via LMS)



Communication

Class, Interchange by FU-HCM LMS, Forum

Discussing actively in your teams and in classrooms

Free to question and answer



Others

Paper notes are required

Off phone

Use laptops under teacher's instruction (No game, no chat in class)

Evaluation Strategy

- Must attend more than 80% of contact hours (if not, not allow to take exam).
- **Evaluating**
 - 02 Progress Test (Q) 2 x 5 %
 - 02 Workshop (Lab) 2 x 5 %
 - 01 Assignment (Prj) 30 %
 - 01 Practical Exam (P) 30 %
 - Final Exam (FE) 20 %
- **Total score**
 - 10% (Q) + 10% (Lab) + 30% (Prj) + 30% (P) + 20% (FE)
- **Pass**
 - Total score ≥ 5 and Final Examination ≥ 4 (of 10)
 - Every components > 0
- **Retake** only the Final Exam when not passed

Assignments/Workshops/Quizzes

Assignments

- Building project in class (**evaluating face to face**) (50%)
- **Some sessions** for social constructivism
- **Contribution** in lectures on our class
- **Mind Maps** of core flow (40%)
- **Enhance Self Worth - Coursera Certification** (10%)

Workshops

- **Contribution** in lectures on our class (participation)
- **Workshops and Peer Reviews in Weeks**

Quizzes

- Use **Safe Exam Browser**
 - **guideline with LMS/exam.fpt.edu.vn**

Assignments/Workshops/Quizzes

▼ Assessment Area



Assignment

Opened: Monday, 5 January 2026, 12:00 AM **Due:** Wednesday, 18 March 2026, 11:00 PM



ProgressTest1

Opens: Monday, 23 February 2026, 7:08 AM **Closes:** Monday, 23 February 2026, 7:35 AM



From Topic 1 to 4: Servlet, Web Container, JDBC and JSP

Not available unless: You belong to **SE1921_SP2026**



ProgressTest1to8

Opens: Monday, 16 March 2026, 7:08 AM **Closes:** Monday, 16 March 2026, 8:05 AM



Not available unless: You belong to **SE1921_SP2026**



Workshop 1st Week

Submissions open: Monday, 5 January 2026, 7:50 AM **Submissions close:** Saturday, 10 January 2026, 11:50 PM **Assessments open:** Sunday, 11 January 2026, 7:50 AM **Assessments close:** Wednesday, 14 January 2026, 11:50 PM



Not available unless any of: You belong to **SE1910_SP2026** ...

Show more ▼



Workshop 2nd Week

Submissions open: Monday, 12 January 2026, 7:50 AM **Submissions close:** Saturday, 17 January 2026, 11:50 PM **Assessments open:** Sunday, 18 January 2026, 7:50 AM **Assessments close:** Wednesday, 21 January 2026, 11:50 PM



Not available unless any of: You belong to **SE1910_SP2026** ...

Show more ▼



Workshop 3rd - 4th Week

Submissions open: Monday, 19 January 2026, 7:50 AM **Submissions close:** Saturday, 31 January 2026, 11:50 PM **Assessments open:** Sunday, 1 February 2026, 7:50 AM **Assessments close:** Wednesday, 4 February 2026, 11:50 PM



Not available unless any of: You belong to **SE1910_SP2026** ...

Show more ▼



Workshop 5th Week

Submissions open: Monday, 2 February 2026, 7:50 AM **Submissions close:** Saturday, 7 February 2026, 11:50 PM **Assessments open:** Sunday, 8 February 2026, 7:50 AM **Assessments close:** Wednesday, 11 February 2026, 11:50 PM



Not available unless any of: You belong to **SE1910_SP2026** ...

Show more ▼

Assignments/Workshops/Quizzes



Workshop 6th - 7th Week

Submissions open: Monday, 23 February 2026, 7:50 AM **Submissions close:** Saturday, 7 March 2026, 11:50 PM **Assessments open:** Sunday, 8 March 2026, 7:50 AM **Assessments close:** Wednesday, 11 March 2026, 11:50 PM

Not available unless any of: You belong to SE1910_SP2026 ...

Show more ▾



Workshop 8th Week

Submissions open: Monday, 9 March 2026, 7:50 AM **Submissions close:** Saturday, 14 March 2026, 11:50 PM **Assessments open:** Sunday, 15 March 2026, 7:50 AM **Assessments close:** Wednesday, 18 March 2026, 11:50 PM

Not available unless any of: You belong to SE1910_SP2026 ...

Show more ▾



Workshop 9th Week

Submissions open: Monday, 16 March 2026, 7:50 AM **Submissions close:** Saturday, 21 March 2026, 11:50 PM **Assessments open:** Sunday, 22 March 2026, 7:50 AM **Assessments close:** Monday, 23 March 2026, 11:50 PM

Not available unless any of: You belong to SE1910_SP2026 ...

Show more ▾



Enhance AI Self Worth Assignment

Opens: Monday, 11 May 2026, 12:00 AM **Due:** Wednesday, 15 July 2026, 12:00 AM

Nộp chứng chỉ để sự ghi nhận năng lực của AI cho OJT . Cách nộp

Claude 101: [Dán Link chứng chỉ]

AI Fluency for Students: [Dán Link chứng chỉ]

AI Fluency: Framework & Fundamental: [Dán Link chứng chỉ]

IBM GenAI Fund" : [Dán Link chứng chỉ specialization]

AI Fluency for Educators: [Dán Link chứng chỉ]

Future Foundation

Enhance AI Self-worth



– Goal

- Have certificate similar this course's content
- Enables to build scalable and accessible enterprise applications

– Over 60 hours (3 courses)

- Claude 101 (<https://anthropic.skilljar.com/claude-101>)
- AI Fluency for Students (<https://anthropic.skilljar.com/ai-fluency-for-students>)
- AI Fluency: Framework & Fundamental (<https://www.anthropic.com/learn/claude-for-you>)
- AI Fluency for Educators (<https://anthropic.skilljar.com/ai-fluency-for-educators>)
- IBM GenAI Fund (1-5) (<https://www.coursera.org/specializations/generative-ai-for-everyone>)



Enhance AI Self Worth Assignment

Opens: Monday, 11 May 2026, 12:00 AM **Due:** Wednesday, 15 July 2026, 12:00 AM

Nộp chứng chỉ để sự ghi nhận năng lực của AI cho OJT . Cách nộp

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AI Fluency for Educators: [Dán Link chứng chỉ]

How to study

- This course is **complex knowledge** (however, it's **attractive and exciting**), so you need to keep tight grip on it
 - **Read**
 - On the books to get the general concept
 - Reference, study, collection from internet, your classmates, forum ...
 - **Attend lectures**
 - Listens, understand, then **make your own notes (required)**
 - Give your explanation about some topic in lectures
 - Ask questions
 - Give some examples that are not existed in your book
 - Practice all the exercises, demo to make your sense
 - **After classes**
 - Discuss your classmate in directly, on forum
 - Do the lab, MindMap, assignments to submit via LMS, and do more exercises
 - Build your teams in yourselves to support together in studying
- **Make question to deposit marks for components in progress**
- **99% attendance participant will be deposited 01 mark in Assignment**



How to exam/test

- This course is **required** following rules, so you **need to focus and practice** your exercises and homework in try your best everyday (**meet deadline and submit on LMS**)
 - **Progress Tests (if students violate, they take 0 marks)**
 - No books and No conversations, must be used SEB
 - **Workshop/Assignment (if students violate, they take 0 marks)**
 - **Obey requirements and lecturers recommended in his lecture**
 - Not copy (copy code, contents, style)
 - Submission of all source code **does not delete anything**
 - **Practical Exam (if students violate, they take 0 marks)**
 - Based on regular of Examination Dept.
 - Some rules and recommended thing in LMS

Academic policy

- Cheating, plagiarism and breach of copyright are serious offenses under this Policy.
 - **Cheating**
 - Cheating during a test, making a project, or an exam is construed as talking, peeking at another student's paper, or any other clandestine method of transmitting information.
 - Cheating during in making lab and assignment as copy source code, copy style, same meaning in progress, ...
 - **To verify, student's code in project can be deleted, then he/she must be retyped to make the program running correctly. Or/And make new required functions (Otherwise, he/she takes 0 marks for his/her project/quiz)**
 - **Plagiarism**
 - Plagiarism is using the work of others without citing it; that is, holding the work of others out as your own work.
 - **To verify, student must be described functionality dataflow and how it work (Otherwise, he/she takes 0 marks for his/her project)**
 - **Breach of Copyright**
 - If you photocopy a textbook without the copyright holder's permission, you violate copyright law.

Enjoy the Course



Be enthusiastic about the material because it is interesting, useful and an important part of your training as a software engineer.



Our job is to help you learn and enjoy the experience.



We will do our best, but we need your help.

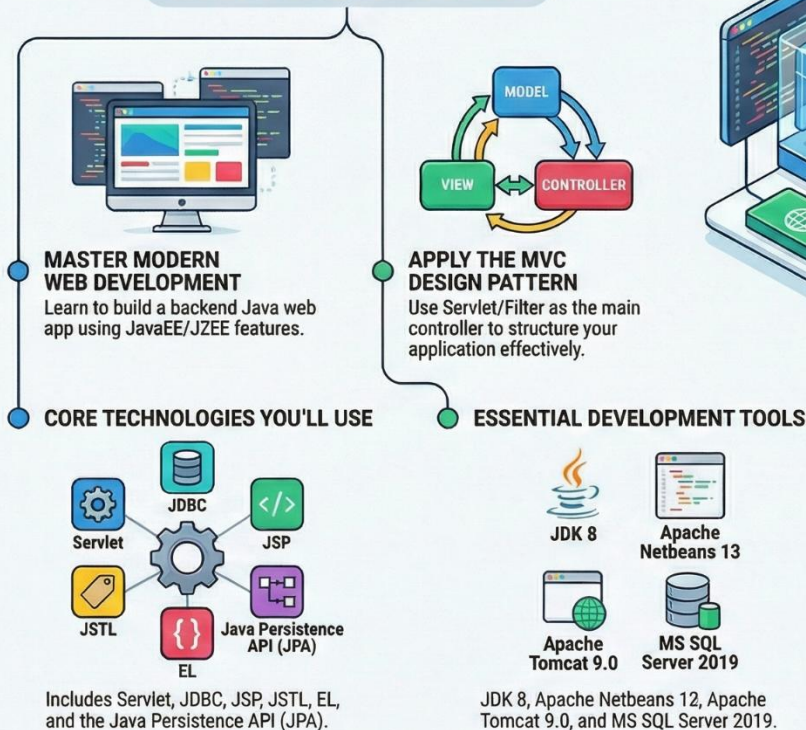


So, let's all have fun together with
Web-Based Java Applications!!!

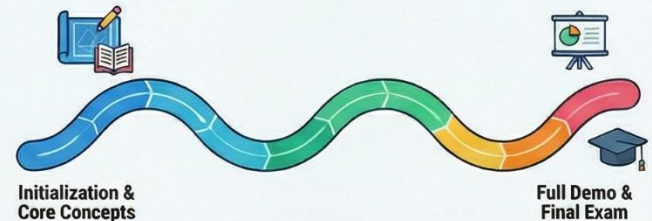
Q & A

A Student's Guide to PRJ301: Java Web Application Development

COURSE FUNDAMENTALS



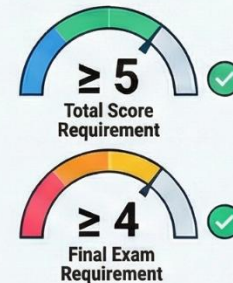
THE 10-WEEK JOURNEY & EVALUATION



A 10-WEEK PROJECT-BASED TIMELINE

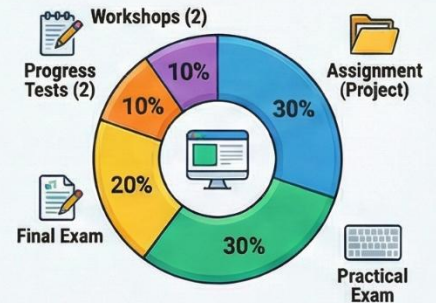
Progress from initialization and core concepts to a full demo and final exam.

PASSING THE COURSE



Requires a total score of ≥ 5 and a Final Exam score of ≥ 4 .

ASSESSMENT COMPONENT WEIGHT



MVC Design Pattern

Remote Controller



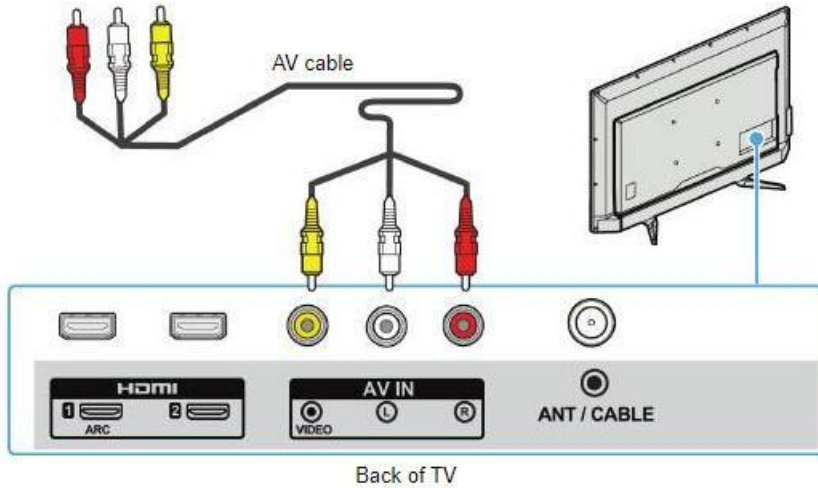
MVC Design Pattern

Player



MVC Design Pattern

Connection



BD/DVD Player



Television

